

Case: Tunnel vision

May 19th, 2026

Executive Summary

We begin by investigating the pcap file discovered over the night from the suspected infected machine. By utilizing zeek we enter in CLI `zeek -Cr` to make several files into more legible files. Next we go into `http.log`, we are going to see what sites were visited using `cat http.log | zeek-cut host | sort | uniq -c | sort -nr`. We see that we had 4 connections to `tunnel.evil-corp.net` and a `C2.evil-corp.net`. After we process uri using `cat http.log | zeek-cut host uri | sort | uniq` to see that we have some executed commands using base64 encoding particularly this command `/?cmd=RkxBR19QVJUMXtETlNfN3VubjNsXw==` which translates to `FLAG_PART1{DNS_7unn3l_` this shows that we found the http tunnel used to execute the commands for the malware on the machine.

Moving forward we run the command `cat files.log | zeek-cut ts tx_hosts filename` to see that there are a few items of interest called `exfil_data.zip` and an `update.exe` file. The `update.exe` is on an IP address of `198.51.100.45` which is suspicious of being our infected machine. We then go into the extract all feature by using `zeek -C -r case_evilcorp_1757852990.pcap policy/frameworks/files/extract-all-files.zeek` to create the `extract_files` directory to examine. We can check seeing some of these come back as our commands from previously stated being executed successfully for malware. We then check our previously mentioned ip with files found to see what went to the IP using `zeek-cut rx_hosts tx_hosts mime_type extracted < files.log | grep "198.51.100.45"` seeing we have one named `extract-1757852990.29589-HTTP-FwCA4U1MF9peGvQMY8`. We then go into the directory of `extract_files` and run command `cat extract-1757852990.29589-HTTP-FwCA4U1MF9peGvQMY8` to see the domain, beacon interval and lastly, our flag part 2 which is `flag_p2 = "D3t3ct3d_2025}"` The malware created a persistence mechanism to run a false windows update from the `update.exe` file previously mentioned. Naming it windows

update and running every five minutes under the hidden executable. Finding the flag of {DNS_7unn3l_D3t3ct3d_2025} finishing the lab and discovering the extracted binary.

Findings and analysis

Finding	Finding Details	Description
Infected Workstation	198.51.100.45	Workstation that received the malware
Tunnel to malicious host	tunnel.evil-corp.net	Tunnel Website that was sending commands to infected computer.
C2 host	c2.evil-corp.net	Host website that infected machine communicated with
Infected binary	extract-1757852990.29589-HTTP-FwCA4U1MF9peGvQMY8	File extracted that shows commands C2 has given as well as our flag.
Flag	{DNS_7unn3l_D3t3ct3d_2025}	Flag after half was decrypted as base64 and half hidden in infected binary.

Tools and technologies used

Zeek – open source network analyzer that creates easy to read logs for analyst.

Investigation process

1. We begin by breaking the pcap file into smaller, easier to manage files by entering `zeek -Cr case_evilcorp_1757852990.pcap` into terminal to break it into `files.log`, `http.log`, `dns.log`, `files.log`, `conn.log` and `packet_filter.log`
2. We then analyze the http file with `cat http.log | zeek-cut host | sort | uniq -c | sort -nr` to see what host were visited, we see there is 4 tunnel connections, a C2 connection and an IP address of 198.51.100.45

```
student@Ubuntu:~/Desktop$ cat http.log | zeek-cut host | sort | uniq -c | sort -nr
 4 tunnel.evil-corp.net
 2 www.microsoft.com
 2 www.apple.com
 1 www.wikipedia.org
 1 www.whitehouse.gov
 1 www.stackoverflow.com
 1 www.noaa.gov
 1 www.nasa.gov
 1 www.mit.edu
 1 www.loc.gov
 1 www.github.com
 1 www.faa.gov
 1 www.cdc.gov
 1 c2.evil-corp.net
 1 198.51.100.45
```

3. Seeing these connections, we want to check the uri request to check for anything suspicious by using command `cat http.log | zeek-cut host uri | sort | uniq`
4. We see normal traffic but we see that the c2 address has an upload, the IP address has an update.exe and that there are 4 separate commands send from tunnel.evil-corp.net with suspicious CLI commands in base64

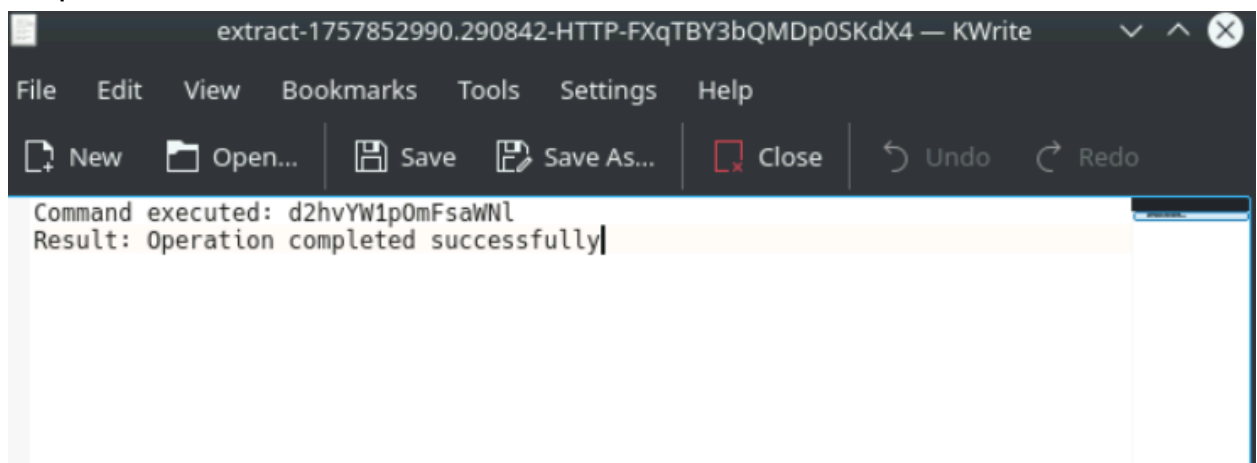
```
student@Ubuntu:~/Desktop$ cat http.log | zeek-cut host uri | sort | uniq
198.51.100.45 /update.exe
c2.evil-corp.net /upload
tunnel.evil-corp.net /?cmd=RkxBR19QVJUMXtETlNfN3VubjNsXw==
tunnel.evil-corp.net /?cmd=aG9zdG5hbWU6REVTS1RPUC1DT1JQ
tunnel.evil-corp.net /?cmd=aXBjb25maWc6MTAuMTAuNTAuNDU=
tunnel.evil-corp.net /?cmd=d2hvYW1pOmFsaWNl
```

5. We run `echo 'RkxBR19QVJUMXtETlNfN3VubjNsXw==' | base64 -d` to see that it decodes to `FLAG_PART1{DNS_7unn3l_` finding the first part of the flag.

6. Next, knowing the c2 host, the tunnel, and the commands executed, we shift into examining the files extracted from the machine as well as confirming the suspicious IP address.
7. We go into the files.log to inspect using `cat files.log | zeek-cut ts tx_hosts filename` tx_hosts filename to see that we have two items of interest. A file called update.exe for 198.51.100.45 and exfil_data.zip

```
student@Ubuntu:~/Desktop$ cat files.log | zeek-cut ts tx_hosts filename
1757852990.266099      23.214.225.151 -
1757852990.268101      23.222.201.22  -
1757852990.271687      23.48.10.36   -
1757852990.274179      192.0.66.108  -
1757852990.277232      104.17.6.58   -
1757852990.280569      3.167.99.25   -
1757852990.283680      23.222.201.22  -
1757852990.286606      23.214.224.163 -
1757852990.288464      23.50.125.163 -
1757852990.290842      198.51.100.47 -
1757852990.292049      198.51.100.47 -
1757852990.293344      198.51.100.47 -
1757852990.294533      198.51.100.47 -
1757852990.295890      198.51.100.45 update.exe
1757852990.297142      10.10.50.45   exfil_data.zip
```

8. Now we want to utilize zeeks extracted file options by using `zeek -C -r case_evilcorp_1757852990.pcap policy/frameworks/files/extract-all-files.zeek` to create an extract_files folder. At a glance, we can look at some of these files to see the commands being entered and the response



9. We dive deeper by sorting any files by specific IP from earlier by using `zeek-cut rx_hosts tx_hosts mime_type extracted < files.log | grep "198.51.100.45"` to see that there is one file associated with this

```
student@Ubuntu:~/Desktop$ zeek-cut rx_hosts tx_hosts mime_type extracted < files.log | g
rep "198.51.100.45"
10.10.50.45      198.51.100.45      application/x-dosexec  extract-1757852990.29589-HTTP-Fw
CA4U1MF9peGvQMY8
```

- [illegible]

1. Enable filtering for only permitted sights to not allow any suspicious uri request to enter commands.
2. Remove or restrict download ability to only trusted domains on end user machines.
3. Add the c2 domain to firewall block list
4. Force traffic through gateways, DNS filtering and proxies
5. Add SIEM detection for Base64 strings or cmd= parameters
6. Enable SSL inspection.
7. Disable or restrict PowerShell and other vectors when they are unnecessary